

Recommender Systems

Lecture 1

Required knowledge / skills

Necessary

Linear Algebra

Coding:

- Python
- Numpy/ Scipy / Pandas
- Jupyter / VS Code / Cursor

Will be helpful

Optimization

Machine Learning

Statistics

Source Control

Kaggle competitions

Course grading

1. Home assignments – 25%
2. Kaggle competition – 25%
3. Mid-term exam – 25%
4. Team project – 25%

Bonus points for extra work (help with bug-fixing, answering difficult questions, etc.)

Kaggle Competition Rules

- several levels worth certain amount of points will be announced
- levels are defined by baseline scores in a **public** leaderboard
 - results below the worst baseline – 0 pts
 - results not worth than the best baseline – 100 pts
- you can get up to 50 pts extra for a top-scoring solution
 - extra points depend on relative position in a **private** leaderboard

Team projects

- 2-5 people per project (single-person projects are not allowed)
- project must prove sufficient efforts towards achieving a goal, e.g.
 - implementing some recsys models (if reproducibility is an issue)
 - performing exploratory research for new ideas
 - developing a recsys application
- you can suggest your own project (requires formal approve)
- you can skip Kaggle and 1 HW without penalty if your project:
 - presents a novel idea outperforming approved baselines and
 - is reported in the form of a high-quality paper draft for an AI-conference

Other requirements

The course is conducted **offline**, presence is required.

Expectations:

- properly annotated charts and graphs in reports
- following python code style guides (e.g., PEP-8)
- vectorized code where possible

Grading:

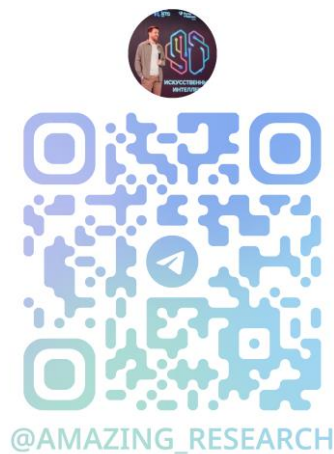
- too inefficient code affects boundary decisions (not in a student's favor)
- code plagiarism is forbidden, will be strictly penalized

Course Instructors

Instructor



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Teaching Assistants



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Reading

Main book:

- **Personalized Machine Learning**, Julian McAuley, 2022;
https://cseweb.ucsd.edu/~jmcauley/pml/pml_book.pdf

Additional reading (in no particular order):

- **Recommender Systems: A Primer**, 2023, P. Castells, D. Jannach, <https://arxiv.org/abs/2302.02579>
- **Recommender Systems Handbook**, 2022, 3rd edition; F. Ricci, L. Rokach, B. Shapira
- **Collaborative Recommendations: Algorithms, Practical Challenges and Applications**; 2019
S. Berkovsky, I. Cantador and D. Tikk.
- **Recommender Systems. The Textbook**, 2016; Charu C. Aggarwal
- **Mining Massive Datasets**, Stanford University, <http://www.mmds.org>
- **Statistical Methods for Recommender Systems**, 2016; Deepak K. Agarwal, Bee-Chung Chen
- **Programming Collective Intelligence**, 2007; Toby Segaran

Helpful resources

Courses

- Julian McAuley's course at UCSD, cse258 class (visit <https://cseweb.ucsd.edu/~jmcauley/>)
- Coursera <https://www.coursera.org/specializations/recommender-systems>

Video tutorials

- Machine Learning Summer School 2014 (by X. Amatriain and D. Agarwal) https://www.youtube.com/playlist?list=PLZSO_6-bSqHQCiyxE3ycGLXHMjK3XV7Iz
- Introduction to Machine Learning 10-701 CMU 2015 (by A. Smola) <https://www.youtube.com/watch?v=gCaOa3W9kM0>

Online resources

- RecSys wiki: <http://recsyswiki.com>
- Fresh RecSys news <https://recommender-systems.com>
- Dive Into Deep Learning – Recommender Systems https://d2l.ai/chapter_recommender-systems/index.html
- Blog: A Practical Guide to Building Recommender Systems <https://buildingrecommenders.wordpress.com>
- Telegram channel (ex-Slack ODS), #recommender_systems (Russian language) https://t.me/ods_recommender_systems

Other sources

Conferences

ACM RecSys

WWW

SIGIR

CIKM

UMAP

KDD

WSDM

IJCAI

...

Competitions

RecSys Challenge

CIKM challenge

Sometimes other conferences, too

Kaggle

...

(Some) Frameworks and Libraries

Polara (*Disclaimer: I'm the author*)

<https://github.com/evfro/polara>

RePlay (Sber AI Lab, supports Spark)

<https://github.com/sberbank-ai-lab/RePlay>

RecTools (MTS)

<https://github.com/MobileTeleSystems/RecTools>

Surprise

<https://github.com/NicolasHug/Surprise>

Turi Create (acquired by Apple)

<https://turi.com/learn/userguide/recommender/introduction.html>

TorchRec

<https://github.com/pytorch/torchrec>

Microsoft Azure

<https://github.com/Microsoft/Recommenders>

Google

<https://www.tensorflow.org/recommenders>

Recbole (modern, PyTorch-based)

<https://recbole.io>

Collaborative Filtering for Implicit Feedback Datasets (iALS /WRMF)

<https://github.com/benfred/implicit> (fastest)

<https://github.com/quora/qmf> (by Quora)

Factorization Machines

<https://github.com/srendle/libfm>

<https://github.com/coreylynch/pyFM>

Other libraries

- Neural Networks

<https://github.com/maciejkula/spotlight>

<https://github.com/MrChrisJohnson/deep-mf>

<https://github.com/songgc/TF-recomm>

<https://github.com/Netflix/vectorflow> (by Netflix)

- Bilinear models

<https://github.com/lyst/lightfm/>

<http://www.recsyswiki.com/wiki/SVDFeature>

- MyMediaLite (used to be popular)

<http://www.mymedialite.net>

- Many latent factor models

<https://github.com/zhangsi/CisRec>

- Logistic-MF (ex-Spotify)

<https://github.com/MrChrisJohnson/logistic-mf>

- Simple content-based recommendation engine

<https://github.com/grovec0/content-engine>

- Hermes (Supports Spark)

<https://github.com/Lab41/hermes>

Recommender Systems Datasets

https://darel13712.github.io/rs_datasets/

Recmetrics – Basic analysis and evaluation

<https://github.com/statisticianinstilettos/recmetrics>

What this course is about

Systematic overview of basic RecSys concepts.

Collection of hints for conducting research and trying ideas.

Practical recipes for using popular recommendation algorithms.

Basic algorithms' implementation and their quality evaluation.

Learning by doing!

Avoid answering wrong questions and solving wrong problems.

"He points out that one of the really tough things is figuring out what questions to ask... Once you figure out the question, then the answer is relatively easy."
Elon Mask's reflections on the "The Hitchhiker's Guide to the Galaxy", by Douglas Adams.

What this course is NOT about

- Building complex RS models (including elaborate Deep Learning / LLM-based solutions) and ensembles
- Distributed systems, map-reduce
- Production-level architectures and pipelines
- Hypothesis verification and user studies

Doesn't mean we won't touch some of these topics.

High-level course structure

Basic concepts

- General problem formulation and basic recommendation techniques.
- Evaluation of recommender systems.

Standard methods

- Collaborative filtering: memory-based and model-based approaches.
- Matrix factorization techniques (a lot of focus on this topic!).

Special methods for collaborative filtering

- Beyond standard MF techniques.
- Closed-form solutions.
- Ranking objectives.

Beyond standard collaborative filtering

- Cold start and hybrid recommender systems.
- Context-aware recommender systems.
- Sequence-aware recommendations.

+ **Kaggle**

+ **Project work hours**

If we'll have time:

- Artificial Neural Networks in RecSys.

What is a recommender system?



Examples:

- Amazon
- Netflix
- Pandora
- Spotify
- Social platforms
- etc.

Many different areas: e-commerce, news, tourism, entertainment, education...

Goal: predict user preferences given some prior information on user behavior.

Personalization levels

- Generic / non-personalized
- Stereotypic
 - Matches by grouping users into certain stereotype categories (e.g., targeting, ads).
- Circumstantial
 - Based on current activity, typically short-term.
- Long-term
 - True personalization based on user history.

In a more general sense

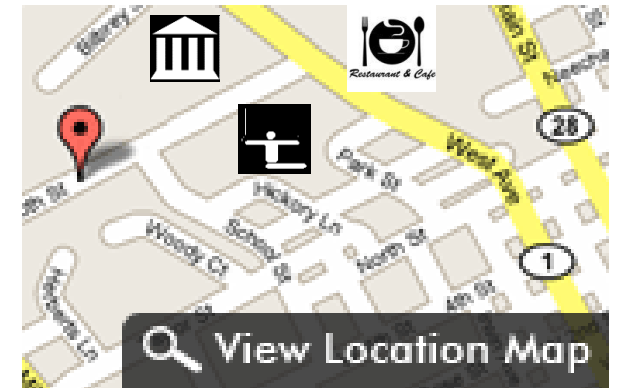
Recommender Systems aim to recover partially observed relations between two or more entities.

Sequential data:
item $\rightarrow$ next item
(order matters)

Social Networks:
user $\leftrightarrow$ user



Ternary relations:
user $\rightarrow$ action $\rightarrow$ location



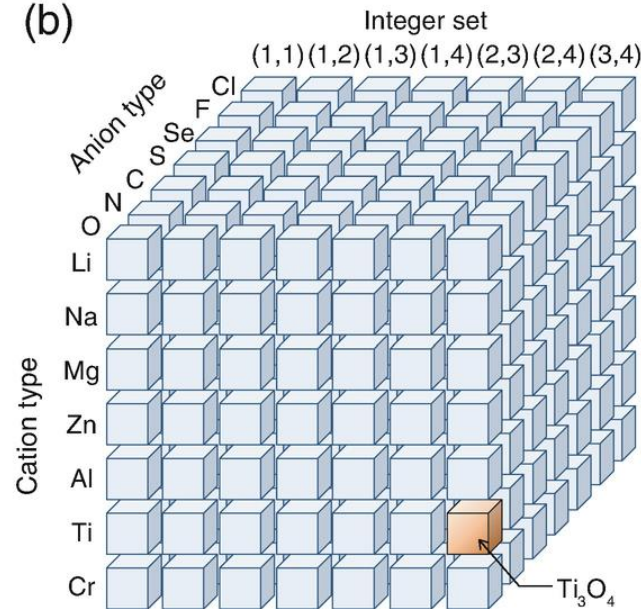
Material Discovery and Recommender Systems

Finding *chemically relevant compositions* and atomic arrangements of *inorganic compounds* using information from inorganic crystal structure databases.

(a)

$$\mathbf{X} = \begin{pmatrix} \text{Li}_2\text{O} & \text{Li}_2\text{O} & \text{Li}_2\text{O} & \text{Li}_2\text{O} & \text{Li}_2\text{O} & \dots & \text{In}_2\text{O}_3 & \text{In}_2\text{O}_3 & \text{In}_2\text{O}_3 & \text{In}_2\text{O}_3 & \text{In}_2\text{O}_3 \\ (1,1,1) & (1,1,2) & (1,2,2) & (1,3,3) & (1,4,3) & & (1,1,2) & (1,1,3) & (1,2,4) & (2,2,5) & (3,3,8) \\ 0 & 0 & 0 & 0 & 0 & \dots & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \dots & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & \dots & 0 & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & 1 & 1 & 1 & 0 & \dots & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & \dots & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & \dots & 0 & 1 & 0 & 0 & 0 \end{pmatrix} \begin{matrix} \text{Na} \\ \text{Mg} \\ \text{Al} \\ \\ \text{Cu} \\ \text{Zn} \\ \text{Ga} \end{matrix}$$

(b)



Discovered compounds and their stability

Composition	Predicted rating	Stability
RbInO ₂	1.01	○
Rb ₃ InO ₃	0.64	○
RbIn ₅ O ₈	0.20	—
NaGaS ₂	0.98	○
NaGa ₅ S ₈	0.21	—
KPbCl ₃	0.97	—
Ca ₂ TiO ₄	0.95	○
CaTi ₃ O ₇	0.29	—
Ca ₃ TiO ₅	0.21	—
BaAs ₂ O ₆	0.93	○
CsZnCl ₃	0.92	○
CsZn ₂ Cl ₅	0.25	○
RbInF ₄	0.91	○
Rb ₃ SbO ₃	0.91	○
CsY ₂ F ₇	0.87	○
Cs ₂ YF ₅	0.44	○
CsYF ₄	0.38	—
CsYS ₂	0.87	○
Ba ₂ Ga ₂ O ₅	0.85	○
BaGa ₄ O ₇	0.55	—
RbZnCl ₃	0.85	○
RbZn ₂ Cl ₅	0.23	—

Image Source

Seko, Atsuto, Hiroyuki Hayashi, Hisashi Kashima, and Isao Tanaka. "Recommender Systems for Materials Discovery." In Machine Learning Meets Quantum Physics, pp. 427-443. Springer, Cham, 2020.

Drug Discovery and Recommender Systems

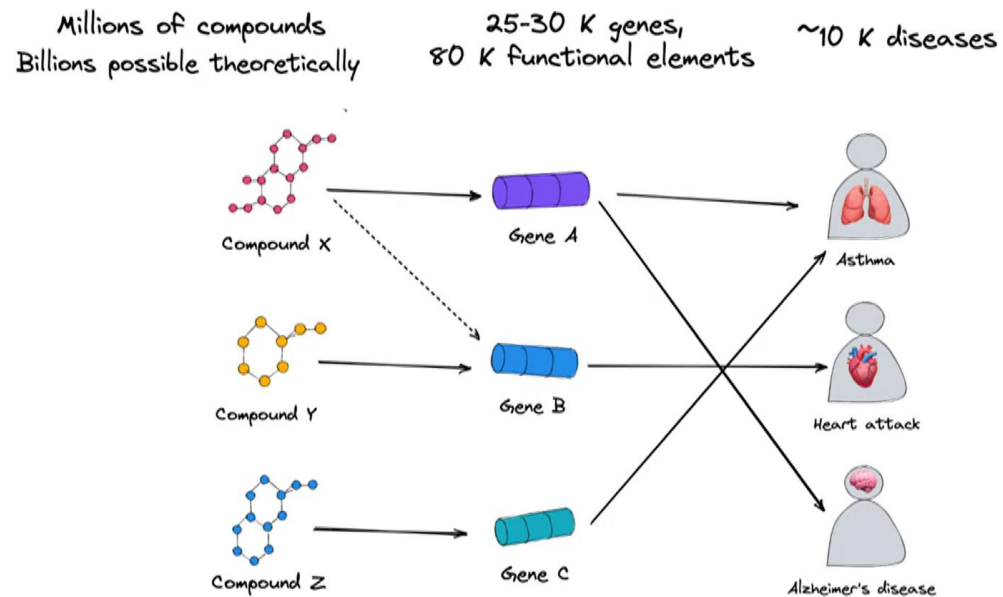


Image Source:

Gogleva, Anna, Eliseo Papa, Erik Jansson, and Greet De Baets. "Drug Discovery as a Recommendation Problem: Challenges and Complexities in Biological Decisions." In *Fifteenth ACM Conference on Recommender Systems*, pp. 548-550. 2021.

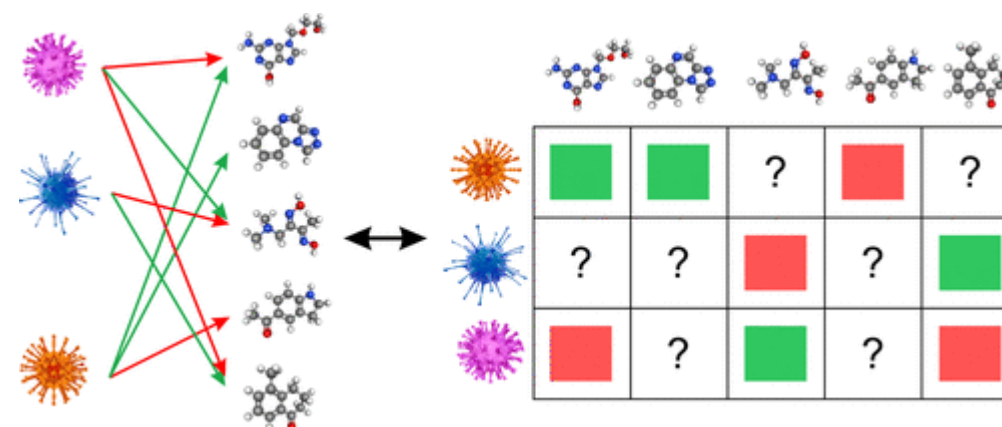


Image Source:

Sosnina, Ekaterina A., Sergey Sosnin, Anastasia A. Nikitina, Ivan Nazarov, Dmitry I. Osolodkin, and Maxim V. Fedorov. "Recommender systems in antiviral drug discovery." *ACS omega* 5, no. 25 (2020): 15039-15051.

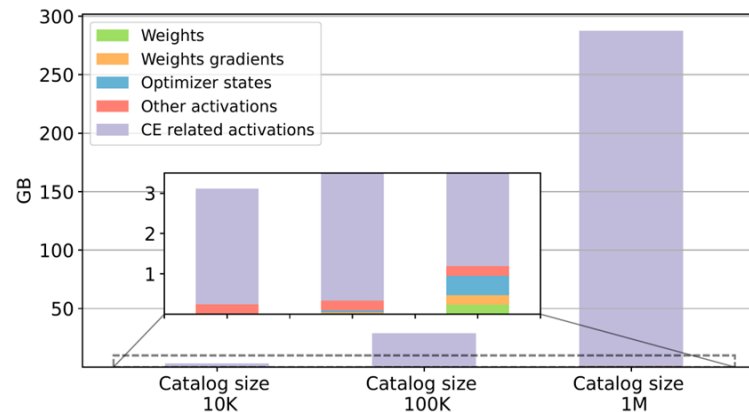
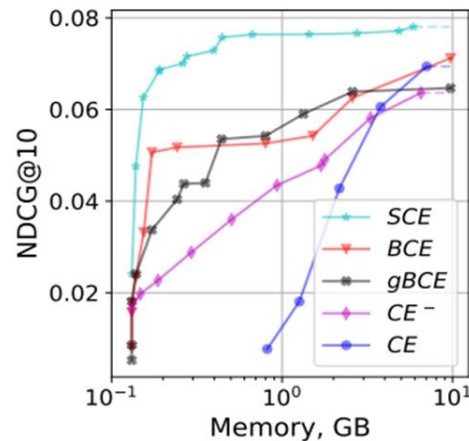
Personalization Technologies Lab (AIRI)

Accurate Recommendations with Hyperbolic Geometry

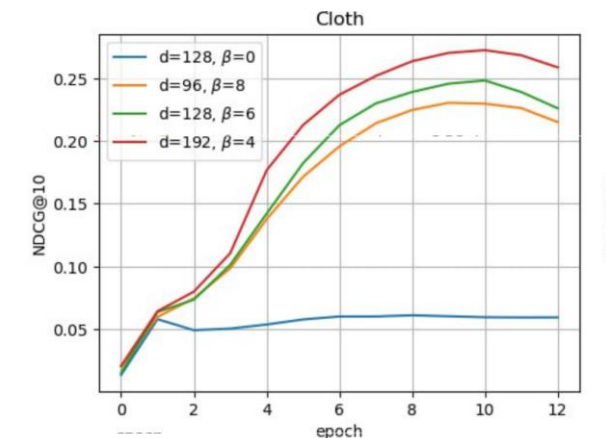
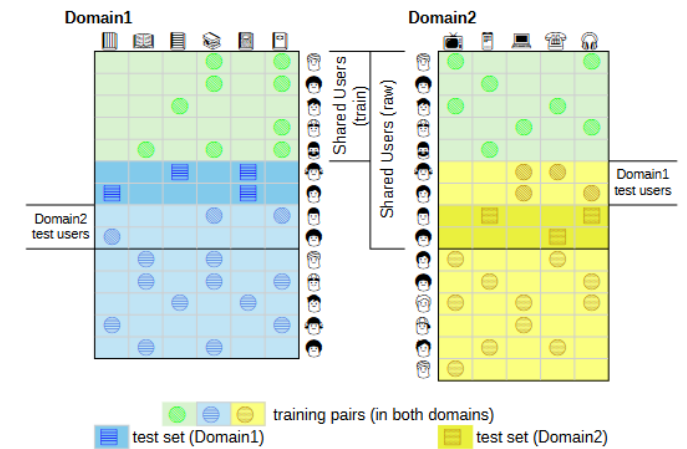


(a) M. C. Escher's Circle Limit III, 1959

Scalable Cross-Entropy Loss for Sequential Recommendations with Large Item Catalogs



Transfer Learning via Cross-Domain Recommendations



Research opportunities

- early research for your profile
- research for diploma
- industrial experience

Rich me if you're interested

Role of personalized services



+\$2.93 billion to revenue after integration of recommendations¹



80% of what people watch comes from recommendations; results in \$1 billion savings²



personalized "Just For You" listings:
“want to be every user’s personalized travel guide”³



“younger travelers ... prefer hotel searches closely tailored to their profiles”, CEO Darren Huston

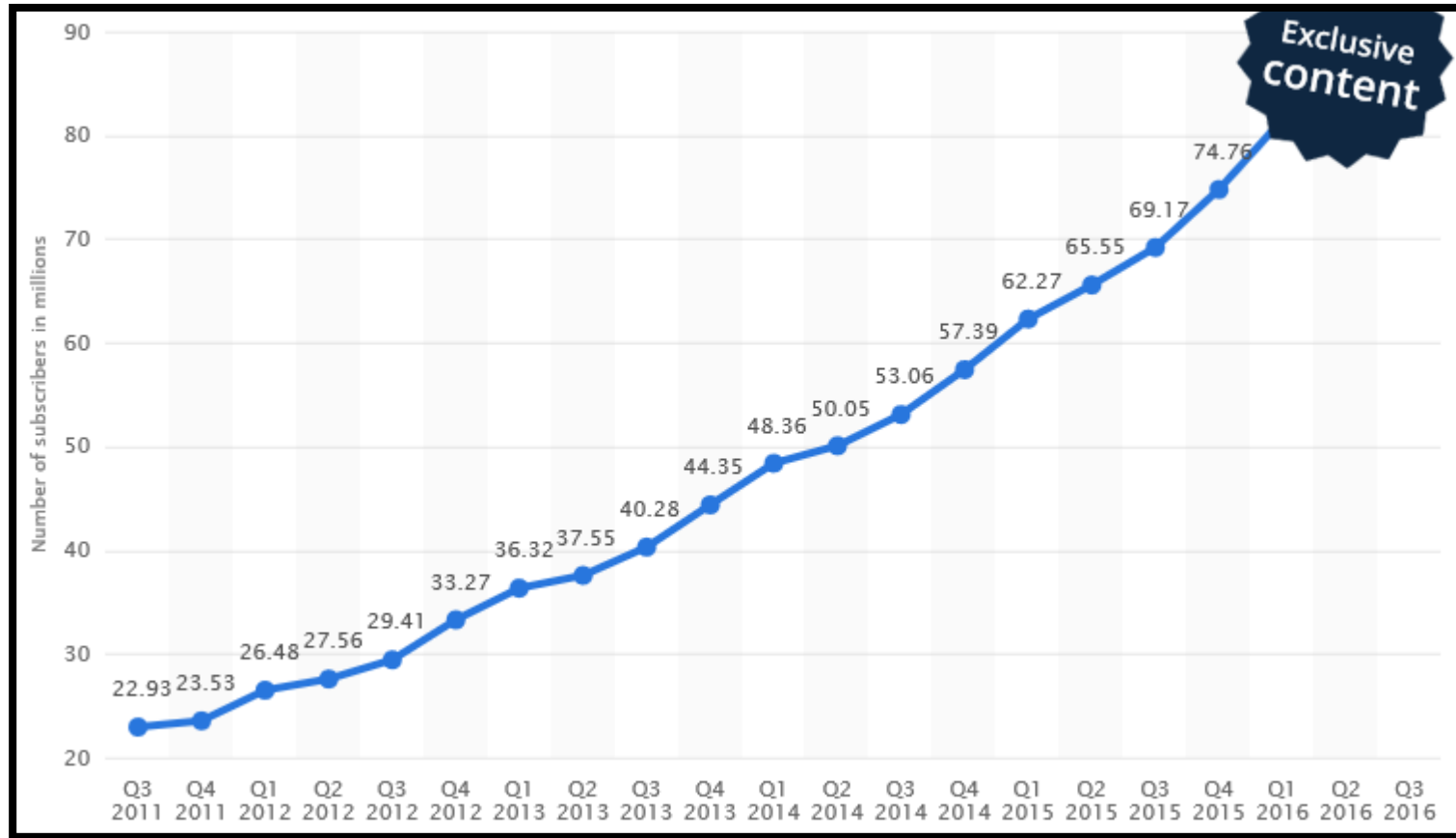
¹ <http://fortune.com/2012/07/30/amazons-recommendation-secret/>

² <http://dl.acm.org/citation.cfm?id=2843948>

³ TripAdvisor’s annual report, april 2015

Example 1

Netflix's audience



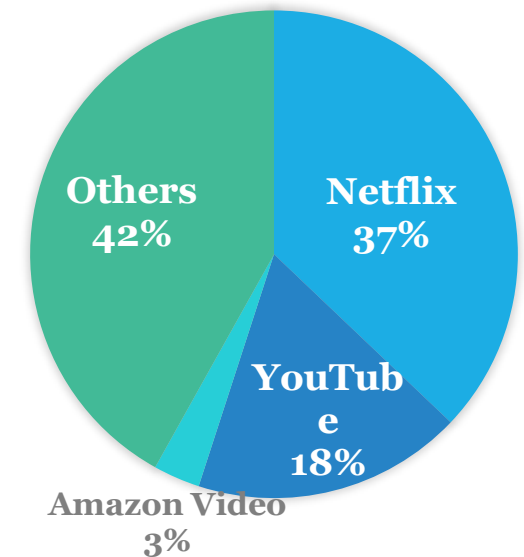
Data sources:

<http://www.internetphenomena.com/tag/amazon-video/>

<https://www.statista.com/statistics/250934/quarterly-number-of-netflix-streaming-subscribers-worldwide/>



Internet video traffic share



Example 2 - IKEA

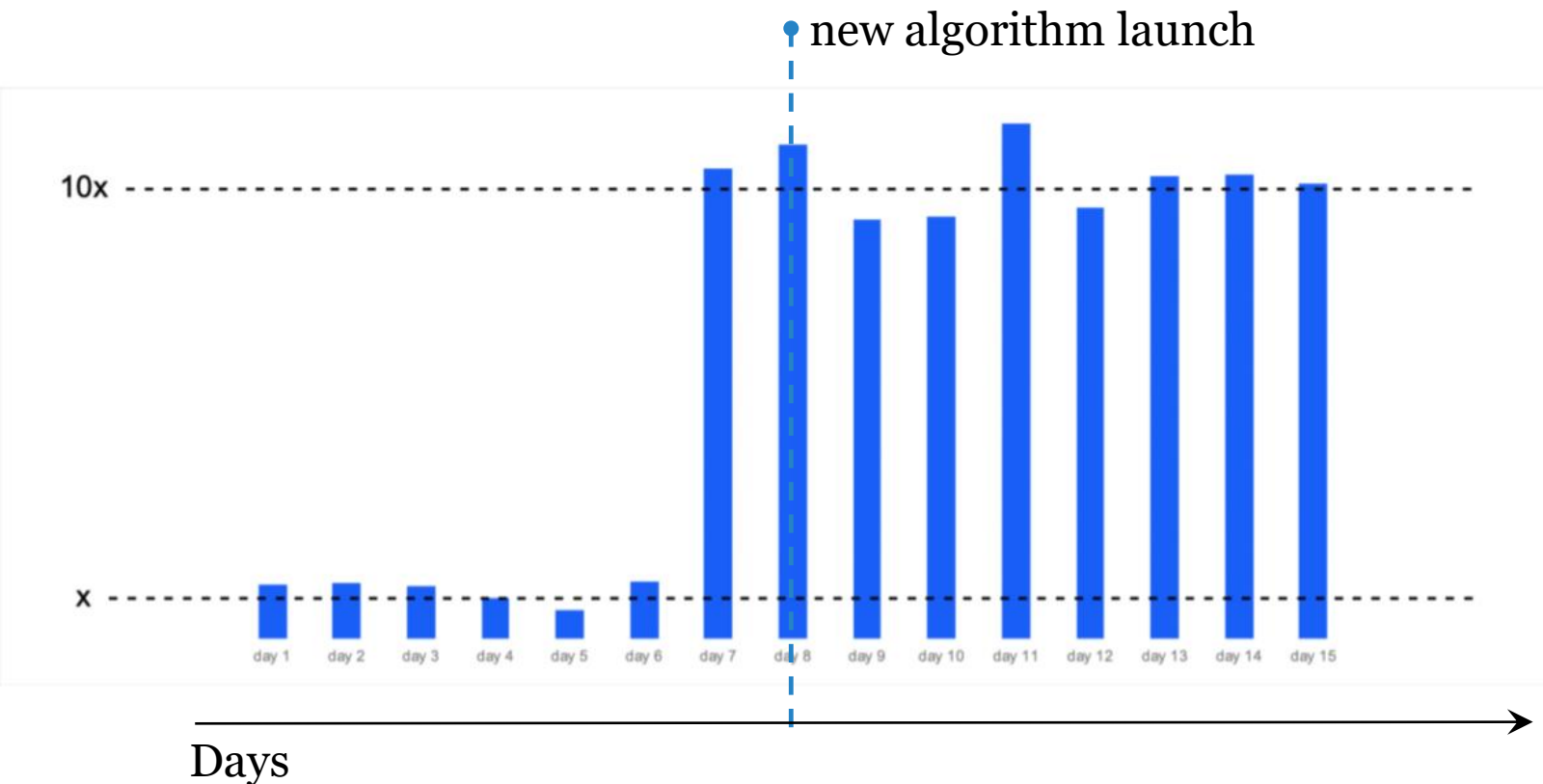
Selling «**inspirational shopping experience**»
based on intelligent matching of products within
a common design style.



Paper: Designer-driven add-to-cart recommendations
<https://dl.acm.org/doi/10.1145/3298689.3346959>

taken from RecSys'19 conference

Example 3 - Ozon



Ozon tries promoting complement products. “*Harry Potter*” problem.

Main target – low CTR products.

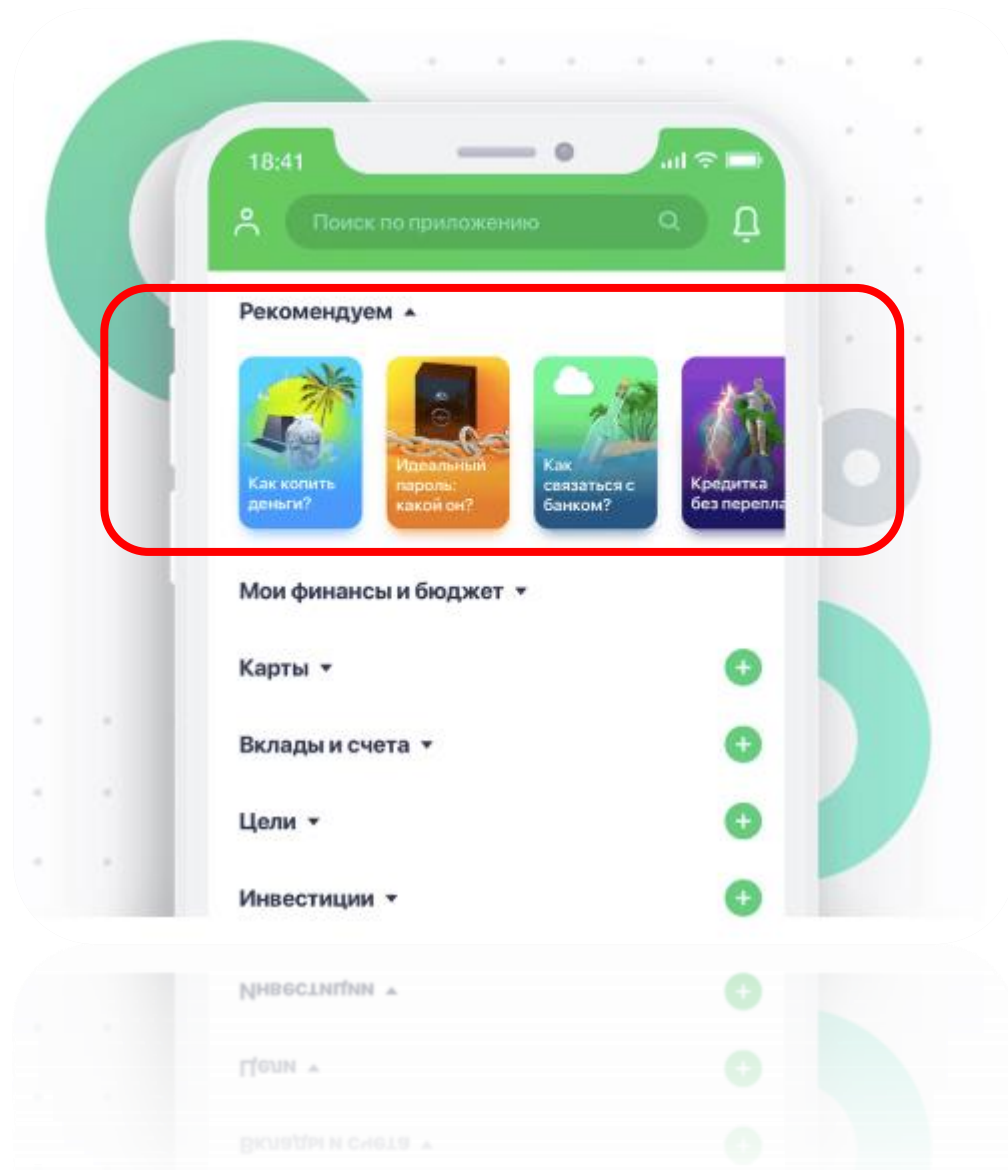


Example 4 – Finance sector

- Individual recommendations of banking products
 - User transactions can be used to guess possible interests, such as:
 - Visit a shopping mall nearby to buy certain products
- What it gives:
 - increases attachment to banking services
 - allows a convenient way of describing possible interests of users
 - improves the connectivity and reliability of the accumulated information about users
 - gives new insights for making business decisions and forming marketing strategy

Such strategies increase revenue in the long-term.

Sberbank Stories Case



Task: displaying relevant “stories” cards.

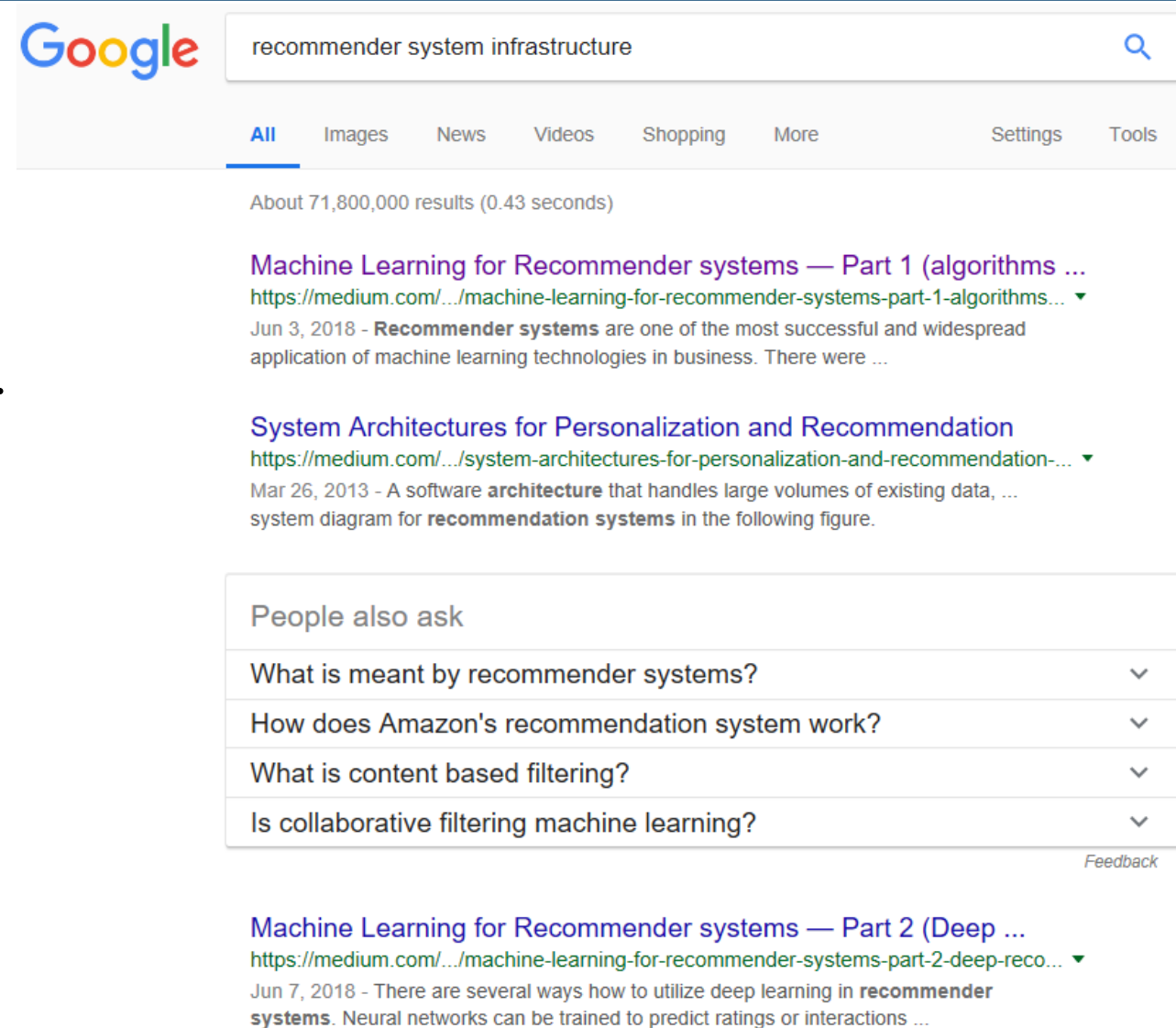
If users actually click them may help with:

- promoting bank's services and products
- spreading news about special offers
 - e.g. with large bonuses / discounts
- stimulating card transactions
 - e.g., ticket sales for movie premieres)

Recommenders Systems vs Search

Search is a query-driven recommender system.

Recommender system is a “query-less” search.



The screenshot shows a Google search interface with the query 'recommender system infrastructure'. The search results page displays the Google logo, the search bar, and navigation tabs for 'All', 'Images', 'News', 'Videos', 'Shopping', 'More', 'Settings', and 'Tools'. Below the search bar, it indicates 'About 71,800,000 results (0.43 seconds)'. The first search result is titled 'Machine Learning for Recommender systems — Part 1 (algorithms ...)' with a URL from medium.com and a date of Jun 3, 2018. The second result is titled 'System Architectures for Personalization and Recommendation' with a URL from medium.com and a date of Mar 26, 2013. Below the search results, there is a 'People also ask' section with four questions: 'What is meant by recommender systems?', 'How does Amazon's recommendation system work?', 'What is content based filtering?', and 'Is collaborative filtering machine learning?'. At the bottom, there is another search result titled 'Machine Learning for Recommender systems — Part 2 (Deep ...)' with a URL from medium.com and a date of Jun 7, 2018. A 'Feedback' link is visible at the bottom right of the search results area.

Google

recommender system infrastructure

All Images News Videos Shopping More Settings Tools

About 71,800,000 results (0.43 seconds)

Machine Learning for Recommender systems — Part 1 (algorithms ...
<https://medium.com/.../machine-learning-for-recommender-systems-part-1-algorithms...>
Jun 3, 2018 - **Recommender systems** are one of the most successful and widespread application of machine learning technologies in business. There were ...

System Architectures for Personalization and Recommendation
<https://medium.com/.../system-architectures-for-personalization-and-recommendation-...>
Mar 26, 2013 - A software **architecture** that handles large volumes of existing data, ... system diagram for **recommendation systems** in the following figure.

People also ask

- What is meant by recommender systems?
- How does Amazon's recommendation system work?
- What is content based filtering?
- Is collaborative filtering machine learning?

Feedback

Machine Learning for Recommender systems — Part 2 (Deep ...
<https://medium.com/.../machine-learning-for-recommender-systems-part-2-deep-reco...>
Jun 7, 2018 - There are several ways how to utilize deep learning in **recommender systems**. Neural networks can be trained to predict ratings or interactions ...

Academia

New, sophisticated, top-performer algorithms

Complexity is not an issue

“Small” datasets

Data-driven verification

Focus on one specific part

Focus on accuracy

vs.

Industry

Good, stable, well-known algorithms

Complexity vs maintenance

Almost unlimited data

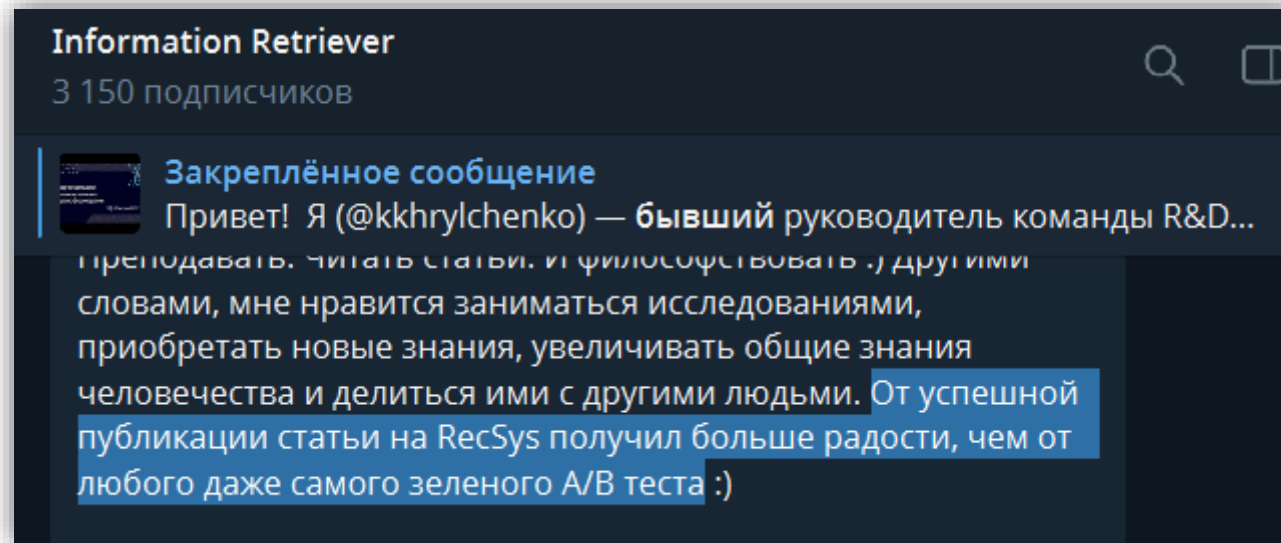
User-driven verification

General view

Balance of various aspects

Trick for productionizing research: read current 3-5 pubs and note the stupid simple thing they all claim to beat, implement that.

Academia vs Industry



A bit of philosophy...

Life is the best recommender and the
person is the Query

<https://www.youtube.com/watch?v=8eBpR9-7swk>

Demo

Simple recommendation engine in 3 lines of python code.

Task for recommender systems

Predict user preferences given some prior information on user behavior.

- What are user preferences?
 - ratings, likes/dislikes, clicks/views
- What is prior information?
 - full user profile, anonymous sessions
- What kind of behavior?
 - short-term, contextual, sequential
- How many users to take into account?
- What kind of prediction is the most appropriate?
 - explanation
 - predicted relevance
- No unified approach (but there're are a few popular ones)
- Many hand-crafted rules and concepts
- Requires creativity
- Requires domain expertise
- A lot depends on user studies
- Good math is preferable but not always required

And...is it really about users in the end?

It is not just for marketing...

"The best minds of my generation are thinking about how to make people click ads"

Jeffrey Hammerbacher

Founder of Cloudera

Source: <https://www.bloomberg.com/news/articles/2011-04-14/this-tech-bubble-is-different>

So what is the main question?

- user satisfaction
- company revenue
- does user satisfaction correlate with revenue?
- what is mathematical formulation?
- how to measure it?

Business goals vs. user satisfaction

Search: Mark Zuckerberg Facebook newsfeed change cost

All Images **News** Videos Shopping More Settings Tools

All news ▾ Past week ▾ Sorted by relevance ▾ Clear

 **Mark Zuckerberg Has Lost This Much Money for Changing ...**
Fortune - 14 hours ago
Facebook stock took a hit after the social network announced massive **changes** to its **news feed**. And no one felt that hit more than **Mark Zuckerberg**. The founder and CEO of **Facebook** owns over 400 million shares of the company, meaning stock fluctuations hit him the hardest. The trick is figuring out exactly how hard ...

Mark Zuckerberg Lost \$3.3 Billion After Announcing **Changes** To ...
HuffPost Canada - Jan 14, 2018

Mark Zuckerberg loses \$3.3bn of personal fortune after **Facebook** ...
Citifmonline - 21 hours ago

Facebook News Feed changes might **cost** the company nearly \$23 ...
BGR India - Jan 14, 2018

I mentored **Mark Zuckerberg**. Here's my road map for fixing **Facebook**.
Opinion - Washington Post - Jan 14, 2018

Facebook's bid to stop fake news could hurt real news outlets
In-Depth - OregonLive.com - 13 hours ago

Over the next few weeks, Facebook's news feed will start showing fewer news articles, and less marketing content and ads, Zuckerberg wrote on Thursday.
<http://fortune.com/2018/01/12/facebook-news-feed-change/>

Typical problems and challenges

cold-start

- recommendation uncertainty
- representative items

missing values

- 99.99...% of unknowns
- subject to biases, Missing Not at Random (MNAR)

short head / long tail

- 5% of items may hold 40% of all interactions)
- niche products

evaluation

- metric choice
- offline evaluation vs. AB-tests

complex models

- incorporating content information
- including context information

performance

- quick model computation
- real-time recommendations

Long tail

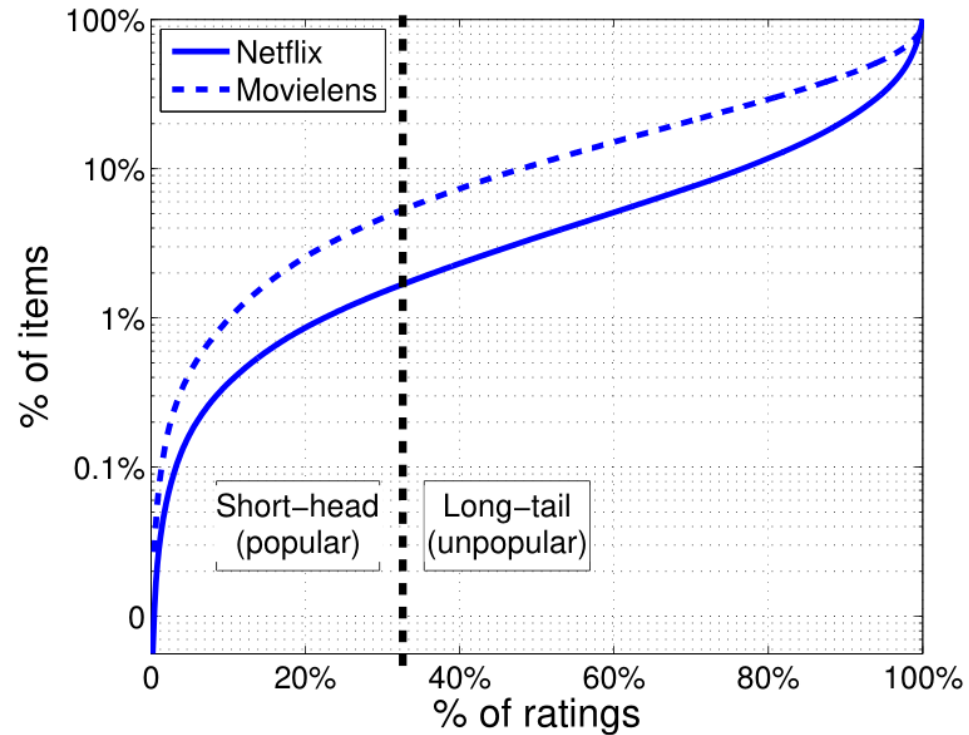


Figure 1: Rating distribution for Netflix (solid line) and Movielens (dashed line). Items are ordered according to popularity (most popular at the bottom).

Image credit: “Performance of Recommender Algorithms on Top-N Recommendation Tasks”, Paolo Cremonesi, Yehuda Koren, Roberto Turrin; ACM RecSys 2010

Missing not at random data - MNAR

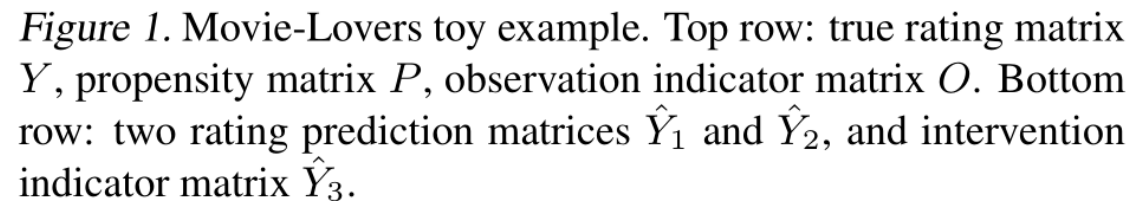
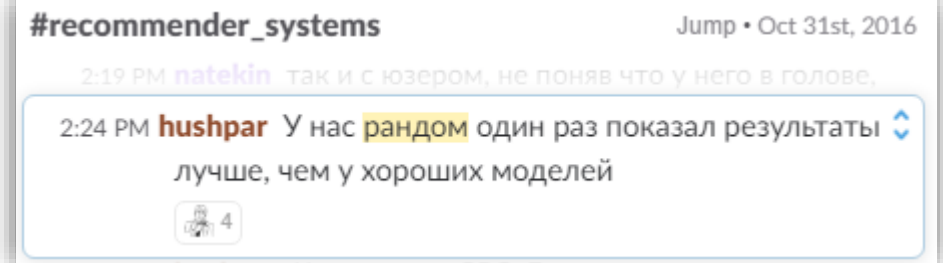


Image credit: “Recommendations as Treatments: Debiasing Learning and Evaluation”,
T. Schnabel, A. Swaminathan, A. Singh, N. Chandak, T. Joachims, 2016

Evaluation difficulties

there're various metrics, but for many of them there's no convenient math formulation

From practical experience:
even when objective's defined, there's still no guarantee that the result will correspond to real RS performance



Translation: Once we had random recommendations outperform our fine-tuned models.

Funny fails

“If you like *The Diversity Myth* book by Peter Thiel you may also like a kettlebell.”

“Need engine parts?
You may find these women shoes relevant.”



[See all 3 images](#)

Follow the Authors

David O. Sacks [+ Follow](#)

Peter Thiel [+ Follow](#)

this book is a compelling insider's tour of a world of speech codes, "dumbed-down" admissions standards and curricula, campus witch hunts, and anti-Western zealotry that masquerades as legitimate scholarly inquiry. Sacks and Thiel use numerous primary sources—the Stanford Daily, class readings, official university publications—to reveal a pattern of politicized classes, housing, budget priorities, and more. They trace the connections between such disparate trends as political correctness, the gender wars, Generation X nihilism, and culture wars, showing how these have played a role in shaping multiculturalism at institutions like Stanford. The authors convincingly show that multiculturalism is not about learning more; it is actually about learning less. They end their comprehensive study by detailing the changes necessary to reverse the tragic disintegration of American universities and restore true academic excellence.

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Categories

All

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Automotive Tools & Supplies

Consumer Electronics

Condition [see all](#)

☐ New (71)

☐ Used (22)

Price

\$ to \$

Format [see all](#)

☒ All Listings (93)

☐ Auction (19)

☐ Buy It Now (74)

Item Location [see all](#)

☒ Default

☐ Within

100 mi of 13159

☐ US Only

☐ North America

☐ Worldwide

Delivery Options [see all](#)

☐ Free shipping

Find parts for your automobile [Go to My Vehicles](#)

Select a vehicle: Select Make Any Model Any Year Any Trim Any Engine

[Update results](#)



****1973 PORSCHE 911 2.4 L USED ENGINE ASSY. (911/91)****

\$4,495.00

Buy It Now

51 watching



Porsche 911 engine 2.4 liter 911S rebuilt 1972 long block (Fits: Porsche)

WE have other 356 & 911 engines in stock / buy cores 2

\$9,900.00

or Best Offer

48 watching



Porsche 911 engine 2.4 liter real 1972 911E rebuilt " Long Block" rare motor

WE have other 356 & 911 engines in stock / buy cores 2

\$9,900.00

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Browse related



Women's 3/4 to 2 Solid Leather Shoes



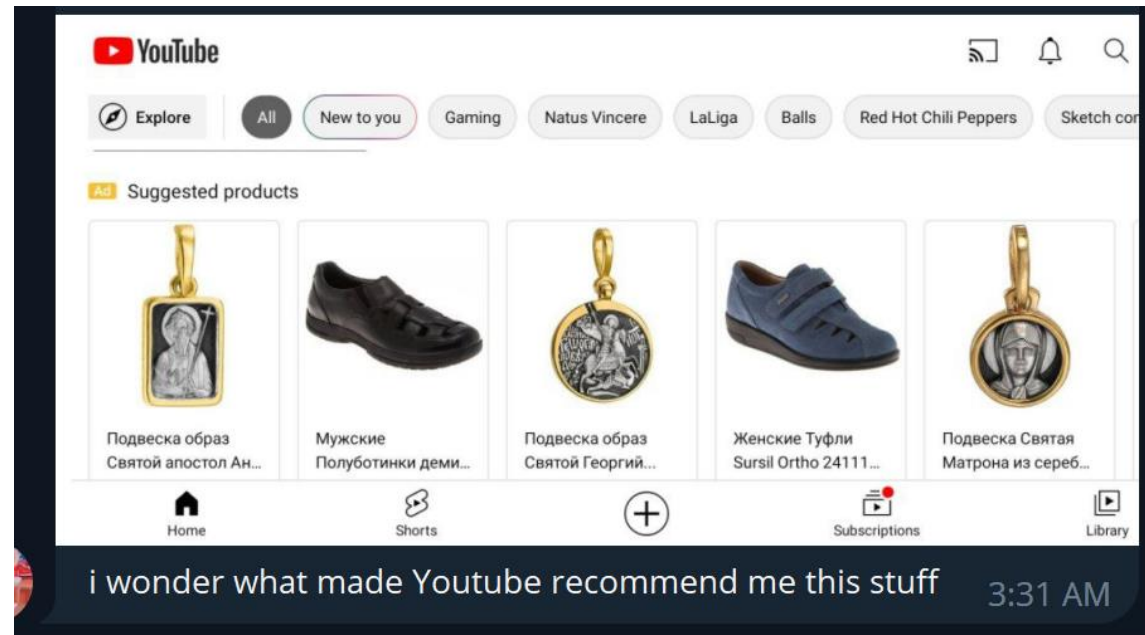
Merona Women's 3/4 to 2 Solid Shoes



Women's 3/4 to 2 Solid Heels

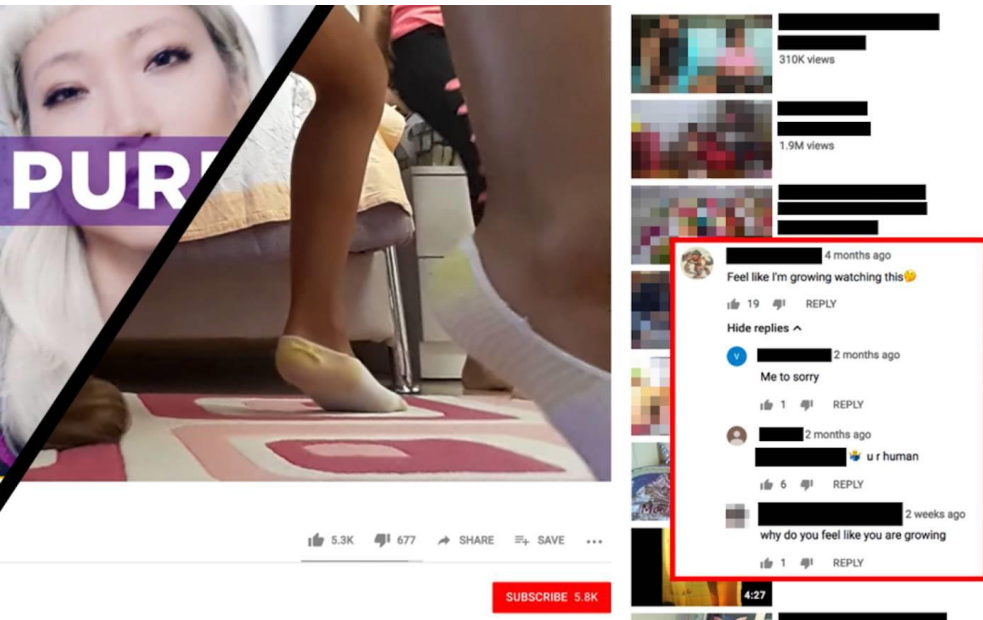
Funny fails

From the previous course participants:



Not funny fails

Unprotected minors and families



<https://www.newsweek.com/youtube-cp-algorithm-ban-mattswatitis-algorithm-1334873>

<https://blog.youtube/news-and-events/an-update-on-our-efforts-to-protect>

Feedback loops

Political radicalization of users



The New York Times @nytimes · Jun 8

Caleb Cain was a college dropout looking for direction. He was then pulled into YouTube's far-right universe, watching thousands of videos filled with conspiracy theories, misogyny and racism. "I was brainwashed."



The Making of a YouTube Radical

Caleb Cain was a college dropout looking for direction. He turned to YouTube, where he was pulled into a world filled with conspiracy theories...

nytimes.com

<https://www.nytimes.com/interactive/2019/06/08/technology/youtube-radical.html>

Algorithms can be biased on variables that aren't part of the dataset. <https://www.fast.ai/2019/05/28/google-nyt-mohan/>

Feedback loops and intervention

Is there a difference between TikTok in the U.S. and China? A social media analyst compares it to opium and spinach

Although they're owned by the same company, China's version of TikTok offers a child-friendly version, with educational videos and a time limit, that isn't offered in the U.S.

<https://www.deseret.com/2022/11/24/23467181/difference-between-tik-tok-in-china-and-the-us>

Related **The content on Chinese Douyin is better than American TikTok. Is this true?**

Yes.

My Tik tok feed (South East Asia) is filled with teenagers doing stupid things.

My Dou yin feed is filled with self improvement, photography, travel, calligraphy, life hacks, trivia, interesting new products, and of course, cute dog/cat videos.

The production quality of douyin videos is generally higher too.

47



1



<https://www.quora.com/Is-it-true-that-TikTok-algorithm-promotes-dance-and-lip-sync-videos-in-the-world-but-scientific-and-engineering-content-in-China>

Suggest possible explanation to these observations.

Filter Bubbles

Network of retweets showing democrats (in blue) and republicans (in red) divided into two distinct communities.

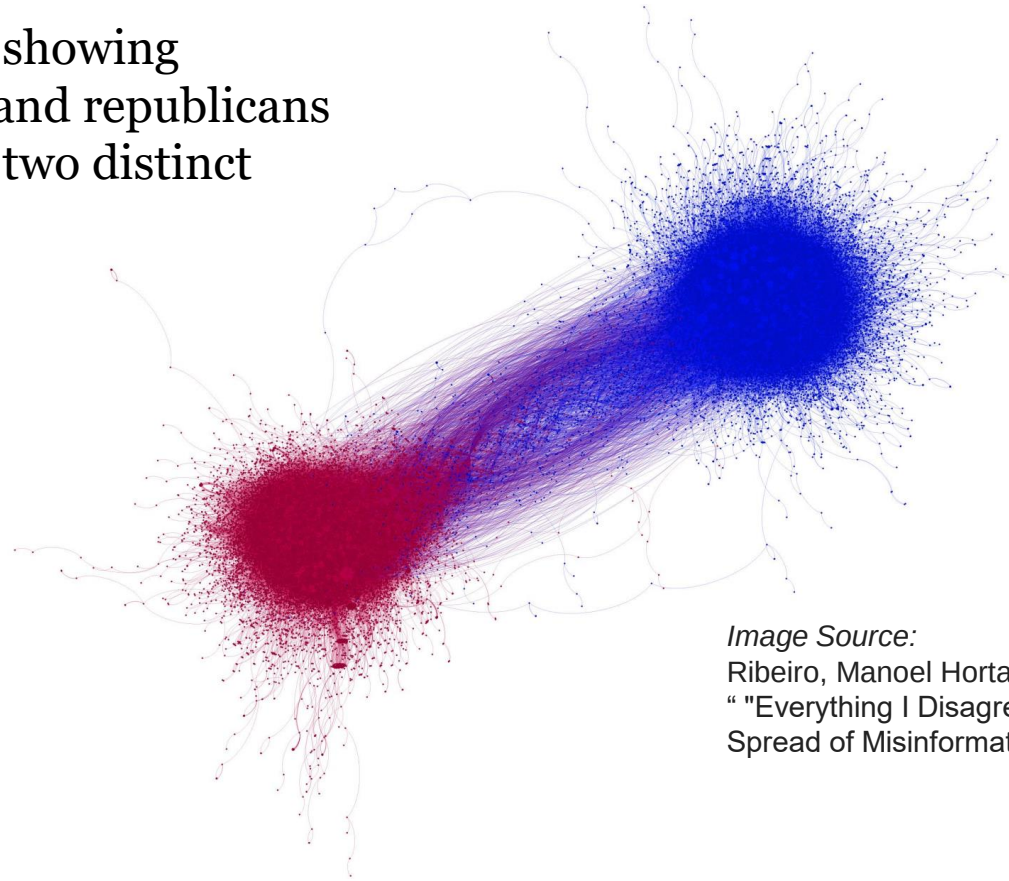


Image Source:

Ribeiro, Manoel Horta, Pedro H. Calais, Virgílio AF Almeida, and Wagner Meira Jr.
“Everything I Disagree With is FakeNews”: Correlating Political Polarization and Spread of Misinformation.”

Additional reading:

<https://www.quora.com/Do-recommender-systems-create-a-“filter-bubble”>

<https://www.quora.com/Should-we-worry-about-the-filter-bubble-created-by-recommender-systems>

<http://open.blogs.nytimes.com/2015/08/11/building-the-next-new-york-times-recommendation-engine/>

Other topics

Multi-task learning

Causality

Federated learning and privacy

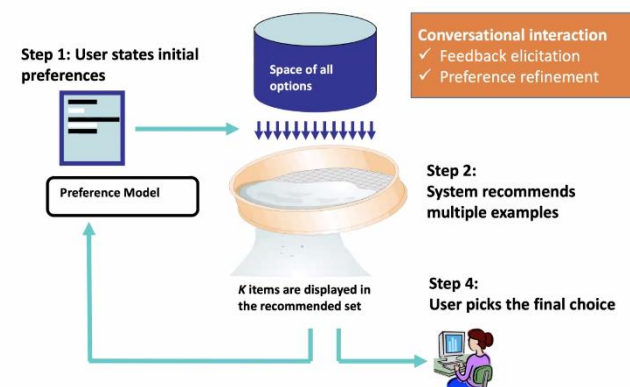
Fairness and debiasing

Reinforcement learning

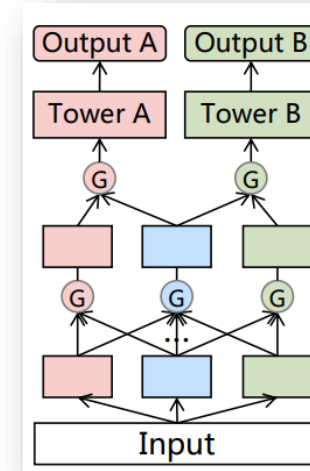
Conversational Recommenders, critiquing

LLM for RecSys, Foundational models, Semantic IDs

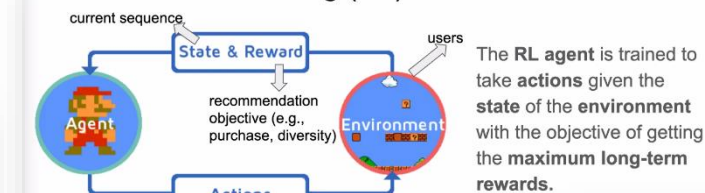
Critiquing-based Recommender Systems



Li Chen and Pearl Pu, Critiquing-based Recommenders: Survey and Emerging Trends, User Modeling and User-Adapted



Reinforcement Learning (RL)

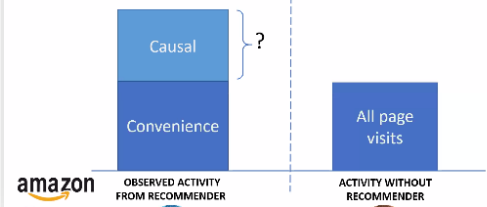


Advantage of RL:

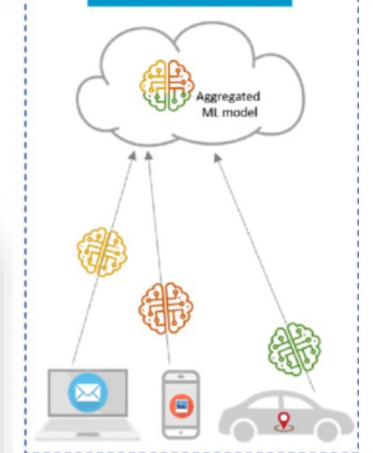
- Flexible reward setting
- Long-term optimization

Casting a recommendation problem

Observed activity is almost surely an overestimate of the causal effect



Federated Learning



Multi-Armed Bandit Problem

Variants!

